1. Introduction to Determinants and Matrices

The video begins by defining what a determinant is and for what type of matrix it can be calculated.

- **Determinant Calculation**: Determinants are exclusively calculated for **square matrices** [09:09]. A square matrix is one where the number of rows (R) equals the number of columns (C) (i.e., R = C), such as 3x3, 2x2, or 1x1 matrices [07:25].
    - **Rectangular Matrices**: Determinants cannot be calculated for rectangular matrices (where R ≠ C), such as 2x3 or 4x3 matrices [08:00].
- **Notation**: Matrices are represented using square brackets (e.g., `[A]`), while determinants are represented using vertical bars (e.g., `|A|` or `det(A)` or `Δ`) [10:09]. The instructor clarifies that although the vertical bars resemble a "modulus" sign, a determinant can be positive, negative, or zero [24:50].

Example:

Consider the following matrices:

1. `A = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]` (3x3 square matrix) - Determinant can be calculated.
2. `B = [[1, 2, 3], [4, 5, 6]]` (2x3 rectangular matrix) - Determinant cannot be calculated.

---

2. Methods for Calculating Determinants

The video details methods for calculating determinants based on the matrix order.

a. 2x2 Matrices

For a 2x2 matrix `A = [[a, b], [c, d]]`, the determinant is calculated as `ad - bc` [14:30].

- **Example**: For `Δ = [[1, 3], [2, 4]]`, the determinant is `(1 * 4) - (3 * 2) = 4 - 6 = -2` [14:34].
- **Number of Terms**: When expanding a 2x2 determinant, there are `2! = 2` terms [15:43].

b. 3x3 Matrices

For 3x3 matrices, the determinant can be calculated by expanding along any row or column, but the video also introduces a "Sarrus' Rule" equivalent for quicker calculation [29:02].

- **General Expansion (Standard Method)**:
    1. Choose any row or column.
    2. For each element a_{ij} in the chosen row/column, multiply it by $(-1)^{(i+j)}$ and the determinant of the submatrix obtained by removing the i-th row and j-th column (this submatrix determinant is called the minor).

3. Sum these products.
   - **Example for `[[1, 2, 4], [2, 1, 2], [-1, 1, 3]]`**: Expanding along the first row:
     - `+1 * det([[1, 2], [1, 3]])` (for element 1)
     - `-2 * det([[2, 2], [-1, 3]])` (for element 2)
     - `+4 * det([[2, 1], [-1, 1]])` (for element 4)
     - The instructor calculates the total number of terms after expansion as `3! = 6` [19:36].

- **Shortcut Method (Similar to Sarrus' Rule)**: This method is specifically for 3x3 matrices and should **not** be used for higher-order matrices (4x4, 5x5, etc.) [38:21].

  1. Write down the matrix elements.
  2. Repeat the first two columns to the right of the matrix.
  3. Multiply the elements along the main diagonals (top-left to bottom-right) and add them.
  4. Multiply the elements along the off-diagonals (bottom-left to top-right) and subtract them.
  - **Example for `A = [[0, 1, 2], [1, 2, 1], [3, 1, 3]]`:**
  - `0 1 2 | 0 1`
  - `1 2 1 | 1 2`
  - `3 1 3 | 3 1`
    - **(Downward multiplications)**: `(0*2*3) + (1*1*3) + (2*1*1) = 0 + 3 + 2 = 5` [31:01]
    - **(Upward multiplications)**: `(3*2*2) + (1*1*0) + (3*1*1) = 12 + 0 + 1 = 13` [31:18]
    - **Determinant**: `5 - 13 = -8` [31:45]

c. Higher Order Matrices (e.g., 4x4, 6x6)

For matrices with order `n x n`, the maximum number of terms after expansion is `n!` [20:24]. The shortcut method described above is **not applicable** for orders greater than 3x3 [38:21].

---

3. Special Types of Determinants

The video introduces specific matrix types where determinant calculation is simplified.

- **Diagonal Matrix**: A square matrix where all non-diagonal elements are zero. The determinant is simply the product of the diagonal elements [23:49].
  - **Example**: For `A = [[1, 0, 0, 0], [0, 2, 0, 0], [0, 0, 3, 0], [0, 0, 0, 4]]`, `det(A) = 1 * 2 * 3 * 4 = 24` [24:20].
- **Upper Triangular Matrix**: A square matrix where all elements below the main diagonal are zero. The determinant is the product of the diagonal elements [27:01].
  - **Example**: For `A = [[1, 2, 3], [0, 4, 5], [0, 0, 6]]`, `det(A) = 1 * 4 * 6 = 24` [27:34].
- **Lower Triangular Matrix**: A square matrix where all elements above the main diagonal are zero. The determinant is the product of the diagonal elements [28:31].

---

4. Concept of Rotation (Specific Pattern for Even Order Matrices)

The instructor introduces a "Concept of Rotation" for a specific pattern in **even-order square matrices** (2x2, 4x4, 6x6, etc.), which is a favorite in GATE exams [39:10].

- **Pattern**: If a matrix follows a rotational pattern with elements like `1` and `α` at the corners and `0`s elsewhere, its determinant can be calculated using a specific formula.
- **For 2x2**: `[[1, α], [α, 1]]` -> `det(A) = 1 - α²` [39:33].
- **For 4x4**: If the matrix follows the rotational pattern (e.g., `[[1, 0, 0, α], [0, 1, α, 0], [0, α, 1, 0], [α, 0, 0, 1]]`), the determinant is `(1 - α²)²` [41:40].
- **General Formula for Even Order (2n x 2n)**: For such a pattern, the determinant will be `(1 - α²)ⁿ`.
- **Restriction**: This pattern and formula are **only valid for even-order matrices** [44:54].

Example:

For a 4x4 matrix with the described rotational pattern, if `det(A) = (1 - β²)²` and is equal to 0, and `β > 0`, then `β = 1` [46:20].

---

5. Minors and Cofactors

The video briefly explains **Minors** and **Cofactors**, which are essential for determinant calculations and inverse matrices.

- **Minor (M_{ij})**: The minor of an element a_{ij} in a matrix is the determinant of the submatrix formed by deleting the i-th row and j-th column [51:01].
- **Cofactor (C_{ij})**: The cofactor of an element a_{ij} is calculated as $(-1)^{(i+j)} * M_{ij}$ [51:32].
  - **Example**: For `A = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]`, to find the cofactor of a_{23} (element 6):
    - Minor M_{23} = `det([[1, 2], [7, 8]]) = (1*8) - (2*7) = 8 - 14 = -6`.
    - Cofactor C_{23} = $(-1)^{(2+3)} * (-6) = (-1)^5 * (-6) = -1 * -6 = 6$.

---

6. Properties of Determinants

Several key properties of determinants are discussed, which simplify calculations and are frequently tested.

1. **Identical or Proportional Rows/Columns**: If any two rows or two columns of a matrix are identical or proportional, its determinant is **zero** [55:12].
   - **Example**: If `R2` is `2 * R1`, then the determinant is 0 [55:52].
2. **Row/Column of Zeros**: If all elements of any row or any column are zero, the determinant is **zero** [56:47].

3. **Interchange of Rows/Columns**: If two rows or two columns are interchanged, the sign of the determinant **changes** [57:47]. Each swap introduces a negative sign.
4. **Determinant of Transpose**: The determinant of a matrix is equal to the determinant of its transpose ($\det(A) = \det(A^T)$) [01:03:33].
5. **Determinant of Inverse**: The determinant of the inverse of a matrix is the reciprocal of the determinant of the original matrix ($\det(A^{-1}) = 1 / \det(A)$) [01:04:04].
6. **Scalar Multiplication**: If a matrix A is multiplied by a scalar k to get kA, its determinant becomes $k^n * \det(A)$, where n is the order of the matrix [01:11:14].
   - **Distinction from Matrices**: When a matrix is multiplied by a scalar, *every* element in the matrix is multiplied by that scalar. However, when a determinant is multiplied by a scalar, only the elements of *one* specific row or *one* specific column are multiplied by that scalar [01:07:42].
   - **Example**: If $\det(A) = \Delta$ for a 3x3 matrix, then $\det(5A) = 5^3 * \Delta = 125\Delta$ [01:13:16].
7. **Row/Column as a Sum**: If any row or column of a determinant can be expressed as a sum of two or more elements, the determinant can be written as the sum of two or more determinants [01:18:00]. This property is very useful for simplifying complex determinants.
   - **Example**: If a row $R_1 = [a_1 + \lambda_1, a_2 + \lambda_2, a_3 + \lambda_3]$ and other rows R_2 and R_3 remain unchanged, then $\det(A)$ can be split into $\det(A_1) + \det(A_2)$ where A_1 has $[a_1, a_2, a_3]$ as its first row and A_2 has $[\lambda_1, \lambda_2, \lambda_3]$ as its first row, with R_2 and R_3 being the same in both.
8. **Row/Column Operations (Invariance)**: The value of a determinant does not change if an operation $R_i \to R_i + kR_j$ (or $C_i \to C_i + kC_j$) is applied [01:23:03]. This means adding a scalar multiple of one row (or column) to another row (or column) will not alter the determinant's value. This is a crucial property for simplifying determinants by introducing zeros.
   - **Individual Operations**: Applying operations one step at a time (e.g., $R1 \to R1 + 2R2$, then in a separate step, $R2 \to R2 + 3R3$) does not change the determinant [01:26:43].
   - **Simultaneous Operations**: When performing multiple operations in one step:
     - A row/column that is *under transformation* cannot be used to transform another row/column in the same step (e.g., $R1 \to R1 + 2R2$ and $R2 \to R2 + 3R1$ is invalid if $R1$ is changing in the same step) [01:31:00].
     - Row and column operations **cannot** be performed simultaneously in the same step [01:34:06].

Understanding these methods and properties is vital for efficient determinant calculation and is a prerequisite for more advanced topics in linear algebra.